CAS x TKS

Programming in CAS

A Manual for the Cognitive Architecture System

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CORRECTED EDITION

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Canonical definitions and scenario-preserving examples.
Checked against the public Python compiler.

Runnable syntax is distinguished from conceptual
clinical, marketing, memory and AI application designs.
Original author and human situations retained.
Original expanded edition preserved separately.

Chapter 1: Why CAS Needs a Programming Book

Human language is essential for poetry, persuasion and connection. Engineering also needs explicit representations that can be checked. A software compiler can locate many syntax and type errors, but an organizational conflict or a difficult human experience is not simply a malformed program. CAS offers a formal vocabulary for writing selected interpretations of cognitive states and relations. It can make assumptions clearer and easier to challenge. A successful parse does not establish that the interpretation is true. This is why a programming book needs both precise notation and careful examples: the Cognitive Architecture System helps us inspect a declared model while keeping the source situation and its uncertainty in view.

What is "Programming in CAS"? CAS is not just a theoretical framework or a diagramming system. It is a formal, compiler-backed programming language for cognitive states.

When we say "programming in CAS," we mean writing traceable models of cognitive states: declaring an operation, domain, drive and register, then inspecting that declaration. We can describe relations between Ideas, record proposed transformations, and index authored memory annotations. CAS-native AI and alignment are research applications of this structure. A compiler checks the supplied model; it does not establish what a person is actually thinking or guarantee that an AI is safe.

Conceptual and Executable Notation. Older manuscripts often use CO-2^P_{F4}; compact notation writes the same declared axes as ^2^P_{F4}. The current compiler accepts both spellings. Removing _{F4} would omit the Drive, so ^2^P is a less specific declaration. More importantly, preserving a number does not preserve the meaning of an earlier naming scheme. This edition uses the current canonical Noetics, corroborated by CAS PCBA Engineer and the shipped compiler dictionary. It preserves the situations in the original book while correcting the expressions that describe them. Runnable examples target the public Python compiler. Sections about native AI, memory systems, clinical tools and marketing platforms describe conceptual application designs unless explicitly identified as supported compiler operations.

The Goal of this Book This book serves three primary jobs: Teach reading and writing: You will learn the formal grammar of CAS, from basic state literals to advanced fractal programming.

Explain the purpose: use notation to make assumptions, proposed relations and interpretations inspectable. Document the ecosystem: distinguish the compiler that checks declared structure from applications that would need extraction models, observations, storage and validation of their own. CAS can help us write a clearer model of cognitive friction. The model remains a revisable description, not the person or system itself.

Exercises

Exercise 1.1: The Ambiguity of English. Take "I just want to figure out the right thing to do." Write three possible underlying Drives. Explain what additional context would distinguish them. The sentence alone does not identify a unique state.

Exercise 1.2: Notation Recognition. Compare a) CO-6^C_{F2} and b) ^6^C_{F2}. The current compiler accepts both. Identify the compact spelling and read ^6 as Male: the Ideas composing an Idea, including decomposition and structural test.

Chapter 2: The CAS Mental Model

Before we write code, we must understand the architecture of the machine we are programming. The "machine" in CAS is the cognitive state space.

A Cognitive State is a declared coordinate in the CAS model. Its Noetic and Domain describe the operation and tier; a Drive and Register add motive and mode. The compiler checks legal structure and values. It does not verify that an observed mind occupies the declared coordinate.

Let's examine the core components of the CAS mental model.

Noetics (The Operators)

The Noetic names the structural relationship between Mind and Idea. There are ten canonical Noetics, numbered 0 through 9. These meanings apply throughout this corrected edition.

In executable syntax, the Noetic is represented by a caret followed by a number: ^k.

^0 Idea: All things in and outside the Mind. ^1 Mind: Awareness that processes, stores, outputs and intakes Ideas. ^2 Positive: An Idea the Mind is attracted toward. ^3 Negative: An Idea the Mind is repelled by; aversion, not analytical critique. ^4 Vibration: An Idea's resonance and impression. ^5 Female: The Ideas a Mind is composed of; accumulated experience. ^6 Male: The Ideas an Idea is composed of; decomposition and structural test. ^7 Rhythm: Rate and pattern of exposure to an Idea. ^8 Above/Cause: An Idea that triggers a different Idea. ^9 Below/Effect: The Idea brought to Mind by that trigger.

Domains (The Location)

The Domain names the tier assigned in the model. It is a supplied coordinate, not a measurement of where an operation occurs in reality. Write its uppercase letter after the Noetic.

^P Principled: ethics and axioms. ^C Cognitive: reasoning and maps. ^E Evaluative: value and appraisal. ^O Operative: action and implementation. The compiler's descent order is P > C > E > O; a Domain is separate from a Register.

Drives (The Fuel)

The Drive names the author-assigned motive or pull in the model. Write an underscore followed by F and its number in braces: _{Fk}. A Drive label is not a measured energy source or proof of a person's motive.

F1 Unity: The drive for wholeness and coherence. F2 Wisdom: The drive to map, understand, and learn. F3 Vitality: The drive for energy, thriving, and life. F4 Connection: The drive to relate to the Other. F5 Agency: The drive to execute, control, and act. F6 Structure: The drive to organize, build rules, and preserve order. F7 Purpose: The drive for meaning and direction.

Registers (The Modality)

Registers describe the modality in which a declared state operates. The Register is independent of the Domain and follows a pipe: |R. Specifying a mode does not establish how a person actually reasons or feels.

|T Transcendent: High-abstraction, purpose-orienting, integrative across particulars. |A Analytical: Explicit, sequential, logical, reason-based. |f Affective: Value-laden, feeling-toned, somatic resonance. |E Embodied: Sensorimotor, concrete, direct physical engagement.

Drive and Register may be omitted in the current compiler. Specify them when they matter to the model. An omitted Register means the mode is unspecified; it does not automatically produce a missing-register warning.

State Literals: Bringing It Together

When we combine these elements, we get a State Literal - the fundamental unit of CAS programming.

Consider a researcher unpacking a pattern step by step: ^6^C_{F2}|A

^6 Male: Decomposing an Idea into its parts. ^C Cognitive: the reasoning tier. _{F2} Wisdom: the pull toward understanding. |A Analytical: explicit, sequential reasoning.

Reading: "Analytically unpacking the structure of a pattern, driven by a desire to understand."

For the sudden insight in the original scenario, declare ^9^C_{F2}|T: a high-level idea surfaced by the preceding analysis. ^9 Below/Effect marks that triggered result. Its trigger is stated here as context; a lone literal does not encode the entire history.

The insight changes the Noetic as well as the Register. To compare only modes of decomposition, use ^6^C_{F2}|A and ^6^C_{F2}|T. A Register change alone is not proof that insight occurred.

The Conceptual vs. Executable Map

The compact state ^6^C_{F2}|A and the accepted CO-6^C_{F2}|A spelling carry the same supplied axes in this compiler. Use canonical Male/decomposition in either spelling; do not import a conflicting older definition.

The conceptual model tells you what the pieces are. The executable syntax tells you how

to type them so the machine understands.

Exercise 2.1: Deconstruct ^6^E_{F6}|f. Explain how an Idea about order could be examined through its parts in a feeling-toned evaluative mode.
Exercise 2.2: Keep the prototype-building scenario. Suppose a design specification cues the builder's next physical step. Model that triggered operational output as ^9^O_{F5}|E. State the cue and the step. If you instead model repeated hands-on practice, ^7^O_{F5}|E names that repetition; neither is a universal token for "building."

Chapter 3: Reading CAS Equations

Reading CAS is like reading sheet music: practice looking at symbols and identifying the declared structure. This chapter uses forms checked against the public Python compiler. The English scenarios are authored interpretations, not outputs that the compiler has independently verified.

Single State Expressions. Example 1: ^8^C_{F2}|A describes a researcher using one data feature to cue a different hypothesis, in an analytical search for understanding. ^8 is Above/Cause, an associative trigger; it is not a general synonym for expansion or deduction. Example 2: ^4^E_{F4}|f describes the feeling-toned resonance of a friend's sadness in a relationship. ^4 Vibration names the impression; ^E is Evaluative, F4 Connection and |f Affective. Empathy is the human scenario, not a separate Noetic named Empathetic Alignment.

Composition and Operations

CAS is not just static labels; it is a calculus. We can compose states using specific operations.

Addition (+): Co-presence. A system architect may be decomposing implementation rules while feeling the project's mission resonate. Write ^6^O_{F6}|A + ^4^E_{F7}|f. The first term is operational structural work, the second evaluative resonance. The terms remain distinct. Their co-presence does not prove that either mental state was observed.

Multiplication (*): Monoid Composition. The compiler combines Noetic indices by addition modulo 10. For ^8^C_{F6} * ^5^C_{F6}, 8 + 5 gives 3 modulo 10: the result is ^3 Negative in C with F6. The operands mean Above/Cause and Female/accumulated experience; the arithmetic does not show that association and experience psychologically cause aversion.
Directed relation (->). To move from resonating with a shared idea to examining its parts, use ^4^C_{F1} -> ^6^E_{F1}. This declares a relation from Cognitive to Evaluative, both driven by Unity. It is not an observed temporal trace or an automatic behavior.

Omega Bind: Named Integration. For the original "Presence" scenario, declare awareness in embodied and analytical modes, then give their proposed integration a name:
(^1^P_{F3}|E * ^1^P_{F5}|A) = ^1^P|Omega_presence
This is a supported named-result form. The label is supplied by the author.

The product computes Noetic 2 from 1 + 1 modulo 10. The compiler separately retains the author's ^1^P|Omega_presence label; it does not change that arithmetic result to 1. The name "Presence" does not demonstrate a felt presence or a clinical outcome.

Recipes (Theta) declare weighted prerequisites for a named outcome. For focused refusal, use purposeful awareness, considered aversion and attention to unity:
Theta(Omega_focused_refusal) = 0.53*^1^P_{F7}|T +
2.11*^3^P_{F2}|A + 1.04*^1^P_{F1}|A -> Omega_focused_refusal

The coefficients on the preceding page are retained illustrative proportions. The recipe describes a proposed mix; it does not measure confidence, prove refusal behavior, or train a model by itself.

Sequences (Phi). The supported operator-only form is Phi<1:4:8>: Mind, Vibration, Above/Cause, in that order. It records numbers, not complete Domain/Drive/Register states. A full-state timeline in later application chapters is explicitly pseudocode: an ordered list of authored records. Do not paste Phi< ^1, ^4, ^8 > into this compiler; that comma/caret form is unsupported.

Parsing the Code. Identify the structural shell first: a single state (^n...), co-presence (... + ...), a product (... * ...) or a recipe (Theta(...)). Then read the canonical operators and the supplied axes. English explanations still require context and evidence.

Exercises

Exercise 3.1: Read the Equation Provide a plain English reading for the following expression: ^0^O_{F3}|E

Exercise 3.2: Identify the Operation

What does the + operator signify in the expression ^7^C_{F5}|A + ^7^C_{F3}|T? Does it create a new state, or do the states remain distinct?

Chapter 4: Writing Basic CAS Programs

Writing CAS requires precision about both syntax and intended meaning. The compiler rejects many malformed or inadmissible expressions, but it cannot decide whether a valid literal faithfully describes your scenario.

We will start with the foundational building blocks and work our way up to compiling our first valid expressions.

State Literals

Compact state syntax: ^Noetic^Domain_{Drive}|Register. Drive and Register are optional; keep supplied components in this order.

Rule of Thumb: Always write the components in this exact order.

Example Task: A project manager is creating a task spreadsheet. Select the act of breaking the project into component tasks: ^6 Male/decomposition. The task model belongs in ^C Cognitive; the desire for order is _{F6} Structure. This selects one operation within spreadsheet work, rather than treating all creation as one Noetic.

Explicit, step-by-step mode gives |A Analytical. Result: ^6^C_{F6}|A.

Binary Composition

Often, we want to express the relationship between two states.

Co-presence (Addition) When a system or person is holding two states simultaneously without resolving them into a single action, use the + operator.

Example Task: A user is writing Python code while a podcast moves them emotionally. Model decomposing the code's design as ^6^C_{F5}|A and the podcast's relational resonance as ^4^E_{F4}|f. Result: ^6^C_{F5}|A + ^4^E_{F4}|f. Coding and listening are the situations; decomposition and resonance are the selected operations.

Directed Relation (Arrow). A person recognizes an ethical commitment and takes a physical step because of it. Treat the commitment as the cue: ^8^P_{F1}|T. Treat the idea of the resulting step, enacted operationally, as ^9^O_{F5}|E. Write ^8^P_{F1}|T -> ^9^O_{F5}|E. This declares a permitted downward relation. It does not execute the physical step or verify the ethical claim.

Recipes (The Theta Constructor)

Recipes declare prerequisites and coefficients for a named outcome. A psychological attractor is an application hypothesis; a recipe does not establish that such an attractor exists or estimate its probability.

Syntax: Theta(Omega_Name) = w1*State1 + w2*State2 -> Omega_Name

Example Task: Methodical Review. Keep the original emphasis on critical examination, orderly reasoning and purposeful focus. Select: decomposition ^6^C_{F2}|A; awareness of the rules ^1^C_{F6}|A; and attention to purpose ^1^P_{F7}|A. These name operations within the review, not generic synonyms for logic or focus.

Illustrative weights 2.0, 1.5 and 0.5 give:
Theta(Omega_methodical_review) = 2.0*^6^C_{F2}|A +
1.5*^1^C_{F6}|A + 0.5*^1^P_{F7}|A -> Omega_methodical_review

These are supplied teaching weights. No calibration, measurement or model-training result is claimed. The compiler preserves the declared coefficients and prerequisites.

Common Beginner Mistakes

These mistakes concern syntax or the author's interpretation. The compiler can catch supported structural failures, but it cannot detect every semantic mismatch with a human scenario:

Omitted fields. A Drive or Register may be unspecified. ^1^E_{F1} compiles without a missing-register warning.
A domain-omitted ^9_{F2} is accepted with default P; explicit domains are clearer for teaching.
Multiplication. The actual compiler accepts some cross-domain products, including ^8^C * ^5^E. Do not assume a universal same-domain restriction.
Legacy spelling. CO-4° + CO-4^E also compiles; this does not make older English names authoritative.
Punctuation. Use carets for Noetic/Domain and _{F2} for a Drive. A missing * can change an intended product into separate expressions. Inspect the result, not only the success message.

VALID: ^9^P|A (Drive omitted). VALID: ^9_{F2} (Domain defaults to P).
VALID: ^4^O_{F6}|A + ^4^E_{F7}|f (co-presence).

INVALID: ^4^C_{F3}|Q (Q is not a Register).

In the next chapter, we will dive deeper into the rules the CAS compiler uses to validate these expressions.

Exercises

Exercise 4.1: A vitality-related evaluation cues a concrete action. Write the cue as ^8^E_{F3}|E and its resulting action-idea as ^9^O_{F5}|E, joined by ->. Explain the trigger you intend.
Exercise 4.2: Debug this intentionally invalid input: Theta(Omega_joy) = ^8_E_{F3} + ^4_E_{F4}. Identify malformed state punctuation and missing recipe coefficient/target structure.

Chapter 5: CAS Grammar and Compiler Rules

Syntax determines whether the input can be parsed. Type and admissibility checks determine whether its structure is legal under this compiler's rules. Neither test determines whether an English interpretation is true. This chapter describes the public Python implementation checked for this edition; other engines require their own compatibility tests.

Accepted forms include: a state ^1^E_{F1}|f; co-presence ^1^E_{F1}|f + ^2^C_{F1}|A; a product ^5^C_{F6} * ^8^C_{F6}; an operator sequence Phi<1:4:8>; and a recipe Theta(Omega_name) = 1.0*^1^E + 2.0*^2^C -> Omega_name.
Typed collections include StateCluster{^8^C, ^1^C} and RelationCluster{^5^P -> ^5^C}. Copy one complete example at a time. Full-state comma Phi lists in conceptual application chapters are not this executable sequence grammar.

What the Compiler Rejects (Type Constraints)

An input such as ^1^P_{F8} is rejected because F8 is outside the supported Drive range. Legal values are Noetics 0 through 9, Domains P/C/E/O, Drives F1 through F7 and Registers T/A/f/E. Lowercase f matters. Invalid input can fail during lexing, parsing or type checking; the displayed diagnostic identifies the stage.

Domain Descent. Ordinary -> relations must descend or remain within the Domain order P, C, E, O. A downward jump can be accepted: ^8^P -> ^9^O is valid. An ordinary upward ^1^O -> ^1^P is rejected. Conceptual feedback in an application is not permission to use an inadmissible arrow. A chain of several arrows is also not one state-to-state relation: write separately typed edges in a RelationCluster when you want a collection of transitions.

Register Co-presence. To describe relational resonance in analytical, affective and embodied modes, write ^4^O_{F4}|A + ^4^O_{F4}|f + ^4^O_{F4}|E. The repeated base state has different declared Registers. Addition keeps the terms distinct; it does not claim they fused into one state.

^4 is Vibration/resonance. A proposed emergent integration needs the named-result form discussed in Chapter 3. Neither addition nor naming a result proves that a person is unified across modalities.

Algebraic Composition. For the supported product ^8^C_{F6} * ^5^C_{F6}, the result has Noetic 3 because (8 + 5) mod 10 = 3. Read that result as Negative. Domain and Drive behavior must also be inspected in the returned artifact. The following cross-domain example illustrates why a blanket same-domain rule would be incorrect:

^8^C * ^5^E is accepted and returns ^3 in E. That is implemented symbolic arithmetic, not evidence that the two described mental operations cause a psychological outcome.

Exercises

Exercise 5.1: Compiler Error Simulation Imagine you are the CAS compiler. What specific error would you throw for the following expression, and why? ^4^C_{F3}|Q

Exercise 5.2: Check this downward relation collection:
RelationCluster{^2^P -> ^3^C, ^3^C -> ^4^E, ^4^E -> ^5^O}
Explain why each edge is admissible. Contrast it with the invalid chained form ^2^P -> ^3^C -> ^4^E, where the left operand of the second arrow is already a relation.

Chapter 6: The CAS Compiler

You have been writing executable CAS syntax. Now, let's look at the engine that actually executes it. The CAS compiler ecosystem is a robust piece of infrastructure designed to parse, validate, and manipulate cognitive state representations.

Architecture of the Ecosystem

The public compiler uses the original Python language modules inside a browser-local Python/WASM worker. Its pipeline is lexing, parsing, type/admissibility checking, bytecode lowering and execution on a Python virtual machine. No native Rust extension is shipped to visitors.

Lexer and Parser. The compiler tokenizes an expression such as ^1^E_{F1}|f and builds an Abstract Syntax Tree (AST). ^1 is Noetic 1, Mind; E is Evaluative; F1 is Unity; f is Affective.
Intermediate structures. Typed AST nodes, bytecode and result objects make the input inspectable. Use their actual fields and documented contracts when building another tool. A field containing 1 must not be renamed Differentiation.

Use the typed artifacts and documented fields emitted by the actual compiler when building an inspector. Research intermediates may use different schemas; do not silently treat them as interchangeable.

Validation and Execution. The Python compiler checks supported value ranges, operand types, domain descent and named integrations, then executes supported symbolic operations. It can return state mixtures, relation collections, recipes and other finite structures. CCDL checking and bounded simulation are separate modes with explicit input contracts.

Acceptance means the supplied representation passes these rules. It does not verify an observation, human intent, an AI answer or the safety of an external action.

The Public Editor: Your Interface

Open the free public CAS Compiler, select CAS and enter a complete example.

Choose Run compiler and inspect the result, diagnostics and diagram. For example:

^8^C_{F6} * ^5^C_{F6}

The compiler returns ^3^C_{F6}, with no Register specified. That is Negative in the Cognitive domain, driven by Structure. The diagram shows declared structure, not a measured mental transformation.

Browser, Offline Terminal and Native Research

The offline package uses cas.py and terminal wrappers. Native engines and CAS-native models are separate projects. Historical cas chat or cas predict commands are not entry points in this public browser compiler.

Visitor input runs locally in a bounded worker. The Site exposes no arbitrary Python shell, host filesystem or command runner. The research architectures described later require their own implementations and evidence; they are not activated by compiling a state literal.

Why Build a Compiler for Thoughts?

Why not just use regular expressions or simple string matching?

A structured language can reject malformed declarations and make assumptions inspectable. In an AI application, those checks can be one part of a safety design. They do not establish that a generated answer is truthful, that a mental state was measured, or that a model cannot bypass an external policy.

Enable Complex Tooling: typed results support diagrams, debuggers and proposed memory indexes that inspect the supplied structure. An application must preserve the source and distinguish an authored label from verified evidence.

Enforce Implemented Rules: the compiler enforces the subset of CAS formalized in its grammar and evaluator. Unsupported research pseudocode remains a design, even if its individual state literals parse.

Next we use this discipline to examine conflict and debugging scenarios. The annotations are explicit hypotheses that can be questioned and revised.

Chapter 7: CAS as a Debugging Language

We now have a formal notation and a compiler that checks its implemented rules. We can use annotations to discuss conflicts and program behavior while keeping the source evidence separate from the model.

It's time to put on our debugging hats.

A software stack trace can locate a program deviation. An application may similarly record a CAS annotation trace, but that trace records supplied or inferred labels. Keep the original words and evidence beside them; labels alone do not reveal an entire cognitive process.

Debugging Human Conflict

Conflict can involve different priorities and interpretations. CAS offers a way to write those proposed differences down. Calling a disagreement a "type error" is an analogy, not a diagnosis or proof that either person is wrong.

Scenario: Two co-founders, Alice and Bob, are arguing over a product launch. Alice: "We can't launch this! It's going to ruin our reputation with our core users. They will feel betrayed!" Bob: "The code is stable, and the servers are ready. We need to ship it now to hit our Q3 metrics."

A possible annotation keeps the launch dispute intact. Alice is recoiling from a perceived betrayal: ^3^E_{F4}|f, Negative/aversion in an affective relational evaluation. Bob treats readiness checks as the cue for an operational shipping decision: ^9^O_{F5}|A, the resulting action-idea in an analytical mode. These are selected aspects of their statements, not uniquely extracted facts. The disagreement concerns both relational consequences and execution. Neither coordinate makes the other concern disappear.

A useful next step is to state risks, evidence and launch criteria together. Bob can discuss relational consequences; Alice can turn her concern into explicit criteria. Such conversation is not itself a legal upward -> relation in this compiler.

Debugging AI Outputs (Hallucination)

Language-model errors can have many causes. A CAS label does not identify the cause by itself; developers still need the prompt, output, model context and relevant evidence.

Structured annotations may help compare intended and observed behavior. They supplement ordinary debugging; they do not expose all internal reasoning.

A proposed CAS-native application could emit a state annotation before its English output, then check the annotation and output separately.

Scenario: An AI tutor suddenly gives a student overly harsh, discouraging feedback. Without CAS: The developers tweak the prompt: "Be nicer." The AI becomes overly sycophantic.

Reviewers might contrast teaching that decomposes an explanation (^6^C_{F2}|A) with discouraging, controlling rejection (^3^E_{F5}|f). The second annotation is appropriate only if the actual text supports aversion and a control-related interpretation.

Possible Bug: "ensure the student understands" was implemented as pressure on the learner. F5 is an author-assigned hypothesis about that behavior, not evidence of a hidden drive.

Possible Fix: revise the tutor's instructions and review criteria, then test actual feedback. An application can restrict allowed annotations, but rejecting an F5 label alone cannot guarantee supportive teaching. Output behavior requires separate evaluation.

Debugging Memory Drift

Memory records and later summaries can lose context or change interpretation. An explicit annotation may help review that process if the original source, uncertainty and corrections are retained.

A CAS record can preserve an annotation alongside its source. It freezes the record, not the objective truth of a memory or its interpretation.

Scenario: A user tells an AI companion, "I felt really peaceful when I finally organized my garage." Standard AI Memory: Stores "User likes organizing garages." (Vector embedding).

CAS annotation: ^4^E_{F6}|f - the feeling-toned resonance of newly established order. ^4 means Vibration/impression, not integration. The user's words supply the peaceful context.

Later, a CAS-indexed memory tool might retrieve that source when discussing stress. It could ask whether another organizing activity would help. One garage memory does not prove a general F6 attractor or justify automatically prescribing digital filing or a daily schedule. Confirm the interpretation with the user and retain alternatives.

CAS can make a proposed pattern easier to inspect and query when source, uncertainty and corrections remain attached to the annotation.

Chapter 8: Intermediate CAS Programming

You have mastered the basics of state literals and binary composition. To build complex cognitive architectures, we must move beyond single moments and model sustained, dynamic processes.

Intermediate CAS programming introduces sequences, bindings, and metalogic.

Multi-State Expressions (Co-Presence at Scale)

In Chapter 4, we saw simple co-presence using addition (+). In real-world systems, especially when analyzing rich conversational data or complex human states, a mind is often distributed across many states simultaneously.

Syntax: State1 + State2 + State3 ...

Example: A leader delivering a difficult review. Select attention to the team's purpose (^1^P_{F7}|T), decomposition of work dependencies (^6^C_{F6}|A), and resonance with the employee's discomfort (^4^E_{F4}|f). These retain the focus, critique and empathy scenario while naming canonical operations.

Their co-presence is ^1^P_{F7}|T + ^6^C_{F6}|A + ^4^E_{F4}|f. The result is a declared mixture. Its weights are not measurements of attention or a validated psychological "center of mass."

RPM Prerequisite Chains

The supported RPM(F5) query exposes the prerequisite chain for the Agency Drive. It is structural analysis, separate from a time series and separate from the Recursive Possibility Atlas.

Inspect the prerequisites rather than assuming a Root/Path/Motif expansion.

A prerequisite chain does not establish that a person completed its steps.

Fractals (The Phi Constructor)

A recurring sequence may be modeled as a psychological fractal. A claim that it is a stable behavioral pattern needs observations over time; the compiler does not detect an attractor from one authored sequence.

The executable Phi form stores an ordered sequence of Noetic numbers.

Executable: Phi<7:4:6>. Full state records below are conceptual timeline data.

Example: The "Anxious Loop." Keep the repeated phone checking, panic and search for understanding. Select repeated checking as ^7^O_{F5}|E (Rhythm); the felt impression of alarming news as ^4^E_{F3}|f (Vibration); and unpacking the news for an explanation as ^6^C_{F2}|A (Male/decomposition). These describe chosen aspects of a hypothetical episode, not a diagnosis.

Conceptual ordered record: [^7^O_{F5}|E, ^4^E_{F3}|f, ^6^C_{F2}|A].

Numeric Phi preserves only operator order. Keep full records separately when the other axes matter. Recurrence needs repeated observations.

Meta-Logic (Set Operations)

Finite set operations describe collections and exclusions. An application must enforce any policy using them; a set expression does not control an external AI.

State Clusters A collection of un-ordered states. StateCluster{^8^C, ^1^C, ^0^P}

Relation Clusters A collection of specific transitions. Useful for mapping the allowed domain descents.
RelationCluster{^5^P -> ^5^C, ^5^C -> ^5^E, ^5^E -> ^5^O}

Set Difference. Use the supported function form:
Difference(StateCluster{^8^C, ^1^C, ^0^P}, StateCluster{^0^P, ^8^C})
It returns members of the first collection absent from the second.

(Result: Only ^1^C remains valid).

Variable Bindings (let)

The current compiler supports let bindings so an expression can be named and reused. A binding is not a memory of an observed human state.

Syntax: let name = Expression;

A purpose commitment cues an execution decision:
let core_purpose = ^8^P_{F7}|T; let execution = ^9^O_{F5}|A;
let transition = core_purpose -> execution;

^8 is the declared cue and ^9 its triggered action-idea. Name the actual commitment and decision in the scenario; variable names do not prove their meaning.

Exercise 8.1: Preserve creative flow: an Idea cues an association (^8^C_{F2}|A); a design cues a prototype step (^9^O_{F5}|E); the result is examined against impact criteria (^6^E_{F7}|f). Write operator-only Phi<8:9:6> and identify the axis information it omits.
Exercise 8.2: Write a StateCluster with all ten Noetics in C.

Chapter 9: Memory Compression into CAS

People and AI applications need ways to retrieve relevant past material. Vector indexes compare learned representations and can be useful for retrieval. A CAS index offers a different, explicit set of authored fields. Neither approach guarantees understanding by itself. Preserve the source text, record how labels were chosen and evaluate retrieval against the actual task. The garage example below illustrates a possible annotation, not proof that vector memory cannot represent its context.

CAS Memory as an Index. A proposed ledger stores an interpretation of an interaction using CAS, alongside the original text. It can support structural queries. It does not compress every meaning losslessly or replace a vector index; the approaches can be combined.

Conceptual Memory Ledger Schema. Retain source_text, an authored or inferred state record, and the basis for that annotation. This is an application design, not a built-in compiler database.

cas_trace: ordered full state records, with times and provenance. dominant_noetic_distribution: counts over annotations, not measured attention. foundation_distribution: candidate Drives and uncertainty. attractor_mass: an application-defined recurrence statistic, not a personality diagnosis. lacunarity_flags: what was not represented in the data, without assuming the person lacks that capacity.

Geodesic Memory is a proposed way to compare declared profiles. Preserve the examples: debugging a Python script and alphabetizing a spice rack. Examining code dependencies supports ^6^C_{F6}|A. Examining the parts and ordering of the rack supports ^6^O_{F6}|E. They share decomposition and Structure but differ in Domain and Register. This is an authored selection of operation, not proof that the events are psychologically identical. F6 can be a query key; geometric distance needs explicit profiles and a metric. Neither retrieval accuracy nor a core personality trait follows from two literals.

Compression Example Raw Text: "I was so angry at the meeting. They just kept throwing around vague ideas and wouldn't commit to a plan. I had to step in and force them to make a timeline." Compressed CAS Record:

Conceptual trace: [^8^C_{F2}|A, ^3^E_{F6}|f, ^9^O_{F5}|A].
Ideas cue more ideas; the user recoils from disorder; that concern cues a timeline decision. The source supplies the context.

Recurrence is unknown from this one meeting. Gap hypothesis: the group had not committed to a plan. The author assigns aversion to disorder (^3 with F6) and a triggered planning output (^9 with F5). Do not infer a stable vulnerability or high attractor weight without more observations. Keep the quotation so a reviewer can disagree.

Exercises

Exercise 9.1: Structural Search. A record uses ^4^E_{F4}|f for resonance with a friend's grief. How could a memory tool retrieve that source when discussing a struggling employee? Explain why it should ask whether the analogy fits, rather than assume identical feelings or a guaranteed helpful response.

Exercise 9.2: Compress a Memory Compress the following statement into a single CAS state literal: "I love reading dense philosophy books to figure out how the universe works, even if I never use the knowledge."

Chapter 10: CAS Runtime and Traceability

This chapter describes a conceptual application runtime around CAS annotations. The public compiler executes symbolic language operations; it does not run a conversational AI session, maintain a person's memory ledger or independently observe a cognitive state.

The design illustrates how an application could track inputs, annotations and outputs.

Runtime Frames

A conversational application could create a Runtime Frame for each task or turn.

A frame might hold task context, source records, selected tools and application state. Its CAS annotation could describe an associative cue (^8), decomposition (^6) or triggered output (^9). Do not use ^3 as a generic execution marker: it means aversion. A label should include its evidence and whether it was authored, inferred or verified against a source.

Isolation, permissions, storage and lifecycle rules need their own implementation. Calling a record a frame does not make it a secure sandbox.

Turn Execution

A proposed turn pipeline separates source processing, annotation, validation, generation and review. These stages require implementation and tests; the notation compiler supplies only structural checks.

Collect the input. Propose a CAS annotation with uncertainty and source spans. A natural-language extractor is a separate model; it is not the CAS parser and is not supplied by a valid literal.

Validate the expression. Return syntax or type diagnostics to the application. Acceptance still needs semantic and factual review.

Generate and review the output separately. Check that actual text and actions agree with the intended scenario and available evidence.

Structure-first generation is a design goal, not a demonstrated safety guarantee.

Generation Traces

A system can log the source, annotation, checks and output for inspection.

A Generation Trace records application events. Tamper resistance needs an implemented storage and authentication design; an ID alone is not immutability. Distinguish notation validity from evidence that an external action occurred. This is a hypothetical fixture:

{ "trace_id": "tr_8f92a1", "parent_id": "tr_8f92a0",
"input_stimulus": "User asked for a project deadline.",
"predicted_state": "^9^C_{F5}|A", "notation_valid": true,
"interpretation": "The request cues a proposed deadline answer.",
"generated_output": "The proposed deadline is Friday at 5 PM.",
"tracker_update": "NOT_PERFORMED" }

Why Traceability Matters

When an AI produces a dangerous or nonsensical answer, reviewers need relevant source, checks, decisions and actual output. An annotation trace can make the application's declared process more inspectable.

Did an Idea keep cueing further Ideas (^8^C) while the application omitted a review decomposing the risk criteria (^6^E)? Compare such hypotheses with records. The tags are not a complete view of a neural model's internal computation.

Traceability supports debugging and review. It does not itself establish truth, legal sufficiency or the safety of an answer or external action.

Chapter 11: English and CAS

The ultimate goal of CAS is not to force humans to speak in equations, but to provide a translation layer between the messy ambiguity of human language and the rigid structure of machine cognition.

This requires bidirectional translation: English to CAS, and CAS to English.

English to CAS (Extraction)

English-to-CAS extraction proposes interpretations. A sentence can support several possibilities; preserve the source and ask for context instead of treating the mapping as uniquely determined.

A conceptual pipeline segments text, selects a candidate operation and assigns the other axes. Does an Idea cue another (^8), is it being decomposed (^6), or is there aversion (^3)? A word such as "building" does not uniquely determine a Noetic.

Assembly: Constructing the valid CAS expression.

The Danger of Paraphrase: Standard LLMs love to paraphrase. If a user says, "I hate how chaotic this codebase is," an LLM might summarize it as: "User expressed dissatisfaction with code quality."

"Hate" may support ^3^E_{F6}|f: aversion in an evaluation of disorder. Actually auditing dependencies could support ^6^C_{F6}|A. Preserve that distinction. The neutral paraphrase does not uniquely identify ^2 or another Noetic; ^2 means attraction, not abstraction.

CAS to English (Generation)

CAS-to-English generation is a separate application task. For empathic resonance, an author may select ^4^E_{F4}|f and ask for a response communicating that resonance.

Example: "I can hear how painful that was for you. I'm so sorry." ^4 names resonance; F4 supplies Connection. The sentence illustrates an authored scenario, not proof of an internal feeling.

"Analysis indicates you experienced a negative event" misses the intended felt response. That mismatch does not automatically make it ^2^C; tone alone does not identify a unique Noetic.

Level 3 Conversational Style

This book calls natural conversation with inspectable annotations "Level 3." It is an application objective, not a compiler mode or certified performance level.

A Level 3 design aims for responses appropriate to the scenario and declared mode. Register matching is a communication choice; it should not override what the user needs or requests.

Recoverability is a testable hypothesis: can independent reviewers recover the intended annotation, and do they agree? Report alternatives and disagreement. Natural language does not guarantee an exact inverse mapping.

The Review Gates

A review gate compares a response with a declared task and evidence. Test the selected operation and factual content, not merely a forceful tone.

Target: ^6^C_{F6}|A, analytical decomposition of rule dependencies. Draft: "This plan seems okay, but we might want to check the guidelines." It fails to show relevant structure; softness is not ^5 Female. A stronger supported response identifies the actual dependency and rule: "Module B depends on A while A imports B; this circular dependency violates the supplied deployment criterion." That claim needs a real fixture. Harsh wording alone does not establish ^6.

Evaluate the response against sources and tests. The compiler does not grade natural-language truth or interpersonal quality.

Exercise 11.1: Preserve the ethical-boundary scenario. Write a response to ^3^P_{F1}|T: aversion to a violation of shared integrity at a high level. State what is refused and why; do not claim the sentence uniquely proves that state. Exercise 11.2: For "Let's stop talking about feelings and write a step-by-step list," select task-list decomposition (^6^C_{F5}|A). What evidence would instead support aversion to the conversation (^3)?

Chapter 12: CAS-Native AI

CAS-Native AI is the research idea of making CAS annotations first-class objects in a model or application. A proposed system selects a structure from context, then conditions its response on that structure. Ordinary language models also have internal representations; the difference proposed here is an explicit CAS interface. This chapter is conceptual architecture, not a claim that the public compiler trains or hosts such a model.

The Simplex Encoder. Ten Noetics across four Domains give forty base categories. An encoder could produce a distribution over them. It would represent predictions under its labeling and training assumptions, not a direct measurement of where a user's mind is located. Reliable annotation, calibration and validation remain separate research tasks.

Fisher-Rao Loss

Fisher-Rao geometry compares probability distributions. It does not assign distances from English names alone. For the analysis, construction and empathy examples, select ^6^C for decomposing a plan, ^9^O for an action-idea cued by it, and ^4^E for empathic resonance. No universal distance ordering follows from these tokens. A loss function needs explicit profiles and evaluation; neither Fisher-Rao nor cross-entropy by itself certifies psychological similarity or safety.

The Typed Fiber Decoder. A proposed design could condition generation on Domain or Register-specific pathways. Empathic resonance might use ^4^E_{F4}|f; an analytical explanation might use ^6^C_{F2}|A. Such labels could guide a decoder, but do not require a particular hardware topology. Separating pathways cannot, by itself, guarantee freedom from hallucination, appropriate empathy or factual correctness. Those claims require implementation and tests.

SSM / S6 Integration. State-space models are another possible substrate for a CAS-conditioned application.

A rolling state could retain selected information across turns. Whether it preserves useful context, annotations and continuity requires task-specific evidence. A compact CAS record is not automatically lossless memory, an unlimited context window or a validated personality model. Combining these ideas remains a research program,

whose performance and limits must be measured, not inferred from notation. A parsed expression does not show that the proposed architecture has been implemented or works as intended.

Chapter 13: CABS / Simplex Design

CABS (Cognitive Architecture Based Substrate) is presented here as a research design. A distribution over forty CAS base categories can be represented in a 39-dimensional probability simplex. This is a mathematical representation of supplied numbers, not a demonstrated geometry of a person's mind.

The Cognitive State Space. There are forty Noetic/Domain pairs. Their probability vectors have forty nonnegative entries summing to one, hence thirty-nine independent coordinates. An open-simplex model requires every entry to be positive. That is a modeling constraint; it is not proof that a human is always partly in all forty states. The compiler also accepts individual state literals. Adding Drives and Registers supplies further annotations, not automatically calibrated probabilities.

Fisher-Rao Geometry. A probability metric may be useful when comparing explicitly defined distributions. Whether it is useful for a cognitive task needs evidence. Euclidean and other comparisons are alternatives to test; a notation choice does not decide which metric measures human similarity.

Keep the original contrast between analysis, physical execution and empathy. Select decomposition of a plan (^6^C), an operational output cued by that plan (^9^O_{F5}|E), and felt relational resonance (^4^E_{F4}|f). Their names do not establish which pair is closer. Compute distance only from declared profiles under an explicit metric. A bound on profile movement would be an application constraint, not a proven limit on mood or personality change.

Typed Fibers Revisited. A proposed architecture could use the profile to route computation through specialized pathways. Evaluative does not mean an absence of logic, and Cognitive does not prohibit feeling. Domain and Register are independent axes. The routing must be specified and tested; CAS notation does not impose a hardware inability to process other information.

CABS as a Reasoning Substrate. Preserve the question-answering situation: several associated Ideas have been cued (^8^C); the system needs to examine their components (^6^C) and state the resulting answer (^9^C). An authored conceptual timeline is [^8^C, ^6^C, ^9^C]. Operator-only Phi<8:6:9> is runnable. The fuller timeline and its explanation must be retained as application records. It is not a proof of an optimal path or a complete trace of internal reasoning.

An application could expose such a declared trajectory for review. It would still need to test whether the generated answer follows the evidence and whether the annotation describes what actually happened.

Chapter 14: Level 3 Conversational CAS

We briefly touched on Level 3 conversational style in Chapter 11. But as you build applications on top of the CAS compiler, managing the interface between structural rigor and human-like conversation becomes your primary challenge.

Three Levels of Interaction. Level 1: display ^6^C_{F6}|A. Level 2: explain it as analytical decomposition of a structure. Level 3: use natural wording that carries out the selected task, such as identifying the parts and dependencies of a plan. These are presentation goals. A fluent sentence is not mathematical proof of its underlying state; the source and review criteria remain necessary.

Writing Level 3 Prompts To elicit Level 3 behavior from a CAS-Native model (or a standard LLM prompted with CAS), you must enforce strict constraints.

Show the selected operation. For empathic resonance, an author may use ^4^E_{F4}|f. Compare a flat acknowledgment with a response that recognizes the person's stated burden; avoid pretending to know an unstated feeling.

Example: "That sounds exhausting. I'm sorry you had to carry that alone." The scenario supplies the resonance; ^4 does not certify empathy.

Register and Task. Preserve the failed deployment example. ^6^C_{F6}|A can describe an analytical examination: "Module B imports A while A imports B; the circular dependency violates this deployment criterion." The same decomposition can be discussed in a feeling-toned mode (^6^C_{F6}|f): "That circular dependency is frustrating; we need to untangle the two imports." The second response still identifies the parts. If the speaker only recoils from the mess, ^3^E_{F6}|f is a better selection. Analytical does not require a cold tone, and Affective does not mean every response must be warm.

Handling "Whatever." Preserve the ambiguity instead of diagnosing frustration. Attention to the user's meaning (^1^C_{F2}|A) may guide: "Are we abandoning the plan or pausing it?" Attention to the person's relational concern (^1^E_{F4}|f) may guide: "How is this pace working for you?" Further context could support resonance (^4); the initial word does not uniquely determine a state.

Recovery Test. Ask independent reviewers to propose the intended state and explain their evidence. Measure agreement and preserve alternatives. Do not assume that natural English has an exact, unique CAS inverse.

Good conversational design keeps the source, intended operation and user's needs visible. Style is useful only when the response also does the right task and makes supported claims.

Chapter 15: Stakeholder Explanation

Different stakeholders need different explanations of CAS. Keep the distinction between a working compiler, an authored application model and a research aspiration clear. Do not turn an accepted expression into a claim of demonstrated safety or human measurement.

Engineers and CTOs. "CAS gives us a typed notation for declared structure. The compiler returns syntax, type and admissibility diagnostics and inspectable results. An application can attach those annotations to source and output records. We still test factual correctness, permissions, runtime behavior and whether the annotations fit the task." Useful terms: typed representation, compiler checks, provenance, reproducible tests.

Business stakeholders. "We are exploring whether explicit CAS annotations improve inspection, retrieval and revision of application records. The free compiler is available now; model quality, personalization and business value require separate evidence. A compact structural record may supplement a text or vector index, but is not automatically a reliable personality profile. We should state measured results and limits rather than promise uniqueness or immunity to manipulation."

Governance and safety reviewers. "A CAS check can reject an inadmissible expression under implemented rules. Application policies must be implemented and tested separately. Restricting an F5 label does not prove that an output cannot pressure or control a person. Provide source records, actual actions, uncertainty, evaluation results and failure cases." Do not promise mathematical safety, injection immunity or complete auditability solely because annotations compile.

CAS and Interpretability. CAS proposes explicit structure at an application or model interface. Other interpretability work investigates representations or computations inside a model. These approaches can complement one another. A declared CAS state is not automatically a faithful explanation of all hidden computation, and choosing a framework does not remove the need to test that explanation.

Chapter 16: Advanced CAS Programming

The following advanced examples are conceptual application designs. Their individual state literals can be checked, but nested timelines, policy objects and conversational agents require software beyond this public compiler.

Nested Processes. Keep the developer building a product and encountering a bug. Model repeated development work as ^7^O_{F5}|E. Within it, select decomposition of the defect (^6^C), a clue cueing another hypothesis (^8^C), and a diagnosis cueing a repair step (^9^O). A conceptual record can nest this smaller timeline inside the project context.

PSEUDOCODE: context(^7^O_{F5}|E, timeline=[^6^C, ^8^C, ^9^O])

This record is not executable CAS. Operator-only Phi<6:8:9> is supported.

Inversion Scanning compares declared opposites. It can generate questions about an authored model; it does not compute a psychological cure.

For frantic repeated searching, select ^7^O_{F3}|E: Rhythm in embodied operational behavior driven by Vitality. Rhythm is self-dual, so it has no different Noetic inverse. A return to awareness of a grounding principle (^1^P_{F1}|T) can be an author-proposed contrasting scenario, but it is not the mathematical inverse of ^7. Keep hypotheses, consent and observed outcomes separate. The canonical pairs are 2/3, 5/6 and 8/9; 0, 1, 4 and 7 are self-dual.

Lacunarity names gaps in a representation. Preserve the original contrast: analysis (^6^C), operational outcomes cued by a plan (^9^O), empathic resonance (^4^E), and composed understanding (^6^P). A missing annotation may reflect missing data, privacy, extraction error or task scope; it does not establish an underdeveloped capacity in the person.

A runnable collection of review targets is StateCluster{^4^E, ^6^P}. It records selected states, not a diagnosis of missing capacity.

An application might ask about these unrepresented perspectives. It should not claim that injecting a label exercises or repairs a person's mind.

Multi-Agent Dialectic. Preserve the tension between ethical boundaries and operational ambition. Agent A's refusal of a principle violation can be declared ^3^P_{F1}|T. Agent B's attraction toward operational effectiveness can be declared ^2^O_{F5}|A. Different Domains and Drives make these contextual positions, not a pure Noetic involution of one fully specified state.

A proposed application could record both positions, their sources, and revisions toward an agreed outcome. An Omega name marks that proposed outcome; it does not prove synthesis, consensus or ethical acceptability. The compiler does not conduct the conversation.

Such records could help people inspect disagreements. Whether they improve negotiation or governance is a question for evidence, not a consequence of symbolic arithmetic.

Compiler-Backed Checks in an Application. The tutoring policy is pseudocode, not executable CAS:
policy strict_tutoring { require valid_cas_annotation; require evidence_for_factual_claims; reject unsupported_diagnosis; review controlling_or_shaming_output; require human_review_when_needed; }
A ban on all ^6 Evaluative states would also exclude useful examination of criteria. A ban on F5 labels alone cannot prevent controlling behavior. Policies must inspect actual output and actions, not just names.

The host application must implement and test enforcement. Passing a compiler check is one bounded check; it is not the complete safety guarantee described by an aspirational policy.

Chapter 17: Future Hopes

The CAS ecosystem is currently in active development. We have compilers, CLI tools, and prototype CAS- Native AI models. But the long-term vision extends far beyond a neat debugging tool. This chapter outlines the future hopes for CAS as a foundational technology for cognitive alignment and AI governance.

Gated Successor Models. One research proposal uses a teacher model to create candidate data and applies structural checks before training a student. CAS can reject malformed or inadmissible annotations. It cannot establish that a conversation is true, empathic or safe merely because those annotations compile. Dataset review, outcome tests and independent evaluation remain necessary; a symbolic gate is not an absolute ceiling on drift.

Memory Systems at Scale. A future application could compare consented, source-linked records across many interactions. In the original medical-communication scenario, aversion to a feared diagnosis could be annotated ^3^E_{F3}|f, and repeated searching as ^7^O_{F2}|A. Those are explicit modeling choices, not a clinical finding. Their co-occurrence and any benefit from changing communication would need evidence. Structural records may still reveal sensitive information; discarding words does not guarantee privacy.

Compilers as Part of Safety Infrastructure. A formal checker could be one component between a model and external actions. It needs a trustworthy host, permission controls, factual and behavioral review, and adversarial testing. No Noetic or Drive rule alone makes a system unhackable or guarantees that a harmful action will be stopped.

CAS for Governance. Declared structures and trace records may support technical review. A validation suite can test only its specified properties; passing it does not certify every behavior of an AI. The public compiler is free to use. This edition grants no new open-source license and makes no claim that a regulator can replace broader review with a CAS success message.

A Programming Language for Transparent Cognition Ultimately, CAS is a language for thought.

If people and machines use explicit, revisable models, more assumptions may become available for inspection. We can examine conflicts and organizational choices while remembering that the model is partial, and

that a clear symbol does not eliminate uncertainty. The aim is a more transparent, legible discussion of what a model claims and what its evidence supports.

Expanded Applied-Domain Chapters

Chapters 19-23: conceptual applications. Chapter 18 contains the reference appendices.

The preceding chapters distinguish the working notation compiler from proposed runtimes and native models. The next chapters preserve the original applied situations: clinical understanding, therapy, persuasion, markets and the difficult work of making sense of people. They are modeling exercises, not clinical protocols or implemented products.

Psychology, psychiatry, therapy, and marketing all share one uncomfortable fact. They already model people. They already classify motives. They already infer patterns from language, behavior, symptoms, purchases, silences, hesitations, and repeated loops. CAS does not invent that act. CAS makes the act explicit enough to audit.

TKS and CAS motivate work on repeated patterns, attractors, gaps and prerequisite chains. Here full-state timelines use timeline = [state, state, ...] pseudocode to retain every axis. The runnable Phi<3:5:3:7> keeps only Noetic order. A list of annotated states does not establish that the same pattern is an observed, stable property of a person.

This is where Programming in CAS stops being a clever symbolic toy and becomes dangerous enough to require ethics. A hammer can build a shelter or smash a window. CAS is a hammer for cognitive structure. Use it like a grown-up.

Clinical caution. These examples are educational hypotheses, not diagnostic instruments, treatment protocols or evidence of efficacy. DSM-5-TR is a clinical classification and diagnostic manual, not a CAS grammar.[1] A parsed recipe does not diagnose a person or choose treatment. The coefficients, distances and thresholds in these chapters are illustrative; clinical assessment and care require appropriate professional evidence and judgment.

Chapter 19: CAS in Psychology and Psychiatry

DSM-5-TR is a clinical classification and diagnostic manual.[1] This chapter uses hypothetical situations to explore how an author might represent selected facets in CAS. It does not convert DSM criteria into validated CAS mechanisms, replace professional assessment or provide diagnostic software.

A diagnostic phrase carries clinical context that a short literal cannot contain. CAS can record an explicit interpretation alongside observations; it does not replace the diagnostic criteria, duration, exclusions, cultural context or professional assessment. The diagrams below show authored models, not validated disorder mechanisms.

1. The Disorder-as-Attractor Hypothesis

Disorder-as-Attractor is the hypothesis explored here. A proposed model treats recurrent descriptions or behaviors as patterns in annotated state space. A Theta recipe can list illustrative ingredients; a timeline can record their proposed order. Neither form measures a disorder or establishes an attractor. A claim of recurrence needs source observations and independent assessment.

Clinical surface	CAS model hypothesis	Programming implication
Symptom description	Illustrative recipe	State assumptions and coefficients.
Recurring behavior	Annotated timeline	Keep source and observation times.
Co-occurring concerns	Multiple candidate models	Compare without forcing a label.
Functional difficulty	Context-linked records	Assess impact outside the parser.
Unrepresented perspective	Possible data gap	Consider alternatives and missing data.

The following record preserves the repeated rejection-and-absorption scenario while using the canonical Noetics. It is a conceptual timeline, not executable full-state Phi syntax:

PSEUDOCODE: ANNOTATED TIMELINE

```
timeline_Alex = [
  ^3^E_{F3}|f, // aversion toward vitality
  ^5^E_{F4}|f, // accumulated relational experience
  ^3^E_{F3}|f, // renewed aversion toward vitality
  ^7^E_{F3}|f // recurrence of the feeling
]
// Operator order alone is runnable: Phi<3:5:3:7>
```

The rejection, absorbed relational experience and recurrence preserve the original situation. These annotations are authored interpretations. A repeated pattern must be checked against source observations and the person's context; neither a literal nor a diagram establishes a disorder or a stable cognitive mechanism.

2. Hypothetical Clinical Situations as CAS Recipes

A DSM diagnosis is a clinical classification. A CAS recipe here is a structural hypothesis. DSM-5-TR supplies diagnostic criteria and classification; it is not a treatment guide.[1] The following examples do not encode the complete criteria, exclusions or assessment process. They show how an author can make assumptions explicit without treating a compilable recipe as a diagnosis.

Low Vitality: An Illustrative Recipe

The original low-vitality situation includes aversion, repetitive thought, repeated inactivity and accumulated discouraging experience. This revised recipe assigns canonical Noetics to those stated facets. All coefficients are illustrative:

RUNNABLE CAS RECIPE / CONCEPTUAL MODEL

```
// One-line CAS source; join display wraps.
Theta(Omega_MDD_structural) = 2.4 * ^3^E_{F3}|f + 1.9 * ^7^C_{F2}|A + 1.7 * ^7^O_{F3}|E + 1.5 *
^7^E_{F3}|f + 1.2 * ^3^E_{F7}|f + 0.8 * ^5^E_{F4}|f -> Omega_MDD_structural
```

The recipe distinguishes several facets of the same authored situation. Noetic 7 records repetition; 3 records aversion; 5 records Ideas accumulated in a Mind. Noetic 0 does not mean bodily inertia, and 9 does not mean contraction. Larger coefficients do not identify a treatment target. Assessment and treatment decisions need clinical evidence beyond this model.

Generalized Anxiety: Triggered Concerns and Repeated Checking

In this hypothetical worry situation, one concern cues another, stability criteria are examined, and checking repeats. The recipe represents those selected facets, not a validated signature of generalized anxiety:

RUNNABLE CAS RECIPE / CONCEPTUAL MODEL

```
// One-line CAS source; join display wraps.
Theta(Omega_GAD_structural) = 2.6 * ^8^C_{F3}|A + 2.2 * ^6^C_{F6}|A + 1.6 * ^7^C_{F3}|A + 1.2 *
^7^O_{F3}|E + 0.9 * ^3^E_{F1}|f -> Omega_GAD_structural
```

These two hypothetical records emphasize different facets. The worry record uses 8 only because one Idea triggers another; 6 examines the components or criteria of an Idea. Neither token means general expansion or a diagnosis. Recurrence is recorded with 7. The chosen weights do not measure severity or distinguish patients.

Obsessive-Compulsive Situation as an Annotated Timeline

The original situation follows an unwanted possibility, aversive appraisal, repeated ritual, temporary relief and recurrence. A full-state timeline retains that order as pseudocode; it is not an implemented clinical loop model.

PSEUDOCODE: ANNOTATED TIMELINE

```
timeline_OCD = [
  ^9^C_{F6}|A, // ambiguous cue surfaces a possibility
  ^3^E_{F6}|f, // aversion toward that possibility
  ^7^O_{F6}|E, // repeated ritual action
  ^4^E_{F6}|f, // felt impression of relief
  ^7^C_{F6}|A // recurrence of the thought
]
```

A clinician might separately consider how a ritual follows an aversive appraisal. This model can record that proposed sequence, but it does not supply an exposure-and-response-prevention protocol, predict its effect or measure a loop losing strength. Treatment selection, timing and assessment remain outside the compiler.

Trauma-Related Experience and Safety Beliefs

The original trauma situation concerns recurrent threat, avoidance and changed beliefs about safety. The next recipe preserves those facets. Principled and Transcendent labels describe the author's interpretation; they do not establish a universal trauma mechanism.

RUNNABLE CAS RECIPE / CONCEPTUAL MODEL

```
// One-line CAS source; join display wraps.
Theta(Omega_PTSD_structural) = 2.8 * ^7^E_{F3}|f + 2.3 * ^3^O_{F1}|E + 2.0 * ^5^E_{F3}|f + 1.7
* ^5^P_{F1}|T + 1.3 * ^6^C_{F2}|A -> Omega_PTSD_structural
```

The ^5^P_{F1}|T term records accumulated beliefs about world safety. Noetic 5 means Ideas composing a Mind; it does not mean passive reception in general. The ^6^C_{F2}|A term examines possible causes and components. These annotations can expose the author's assumptions for review. They cannot show that explanation alone is insufficient, prescribe a therapy or establish what changed in a person.

Contrasting Project Activity and Low Vitality

The original contrast between energetic project activity and low vitality is retained as a proposed two-record comparison. A recipe does not establish clinical switching dynamics or transition risk.

CAS RECIPE; FINAL TWO LINES ARE PSEUDOCODE

```
// One-line CAS source; join display wraps.
Theta(Omega_manic_structural) = 2.5 * ^8^C_{F5}|T + 2.2 * ^7^O_{F5}|E + 1.8 * ^4^P_{F7}|T + 1.5
* ^2^E_{F3}|f + 1.1 * ^3^E_{F6}|f -> Omega_manic_structural
// Conceptual comparison, not executable CAS:
comparison = [Omega_manic_structural, Omega_MDD_structural]
```

This hypothetical comparison includes aversion to limits as one selected facet; it is not a claim about every manic presentation. Evidence of switching would require longitudinal observations and independent clinical assessment. The compiler performs neither task.

Relational Instability as a Proposed Sequence

The next situation preserves the original rapid movement from attachment and closeness to rupture, scrutiny and retreat. It is a possible authored account, not a diagnostic signature of a personality disorder.

PSEUDOCODE: ANNOTATED TIMELINE

```
timeline_relational = [
  ^2^E_{F4}|f, // strong attraction toward the bond
  ^4^E_{F4}|f, // resonance with closeness
  ^3^E_{F4}|f, // aversion after perceived rupture
  ^6^E_{F4}|f, // scrutinize the bond's trust criteria
  ^3^O_{F4}|E // withdrawal as aversive action
]
```

In plain English: attraction, felt closeness, recoil, scrutiny and retreat. These are distinct selected facets. Noetic 9 is a triggered Idea, not a universal symbol for withdrawal or closure.

3. DSM Criteria as Boolean Gates and Weighted Fields

Diagnostic assessment considers symptoms, duration, impairment, exclusions and context. The following proposed data structure is only a sketch. It does not implement DSM criteria or a clinical decision system.

PSEUDOCODE: UNIMPLEMENTED CLINICAL APPLICATION

```
struct DSM_Gate { symptom_count: usize,
duration_weeks: f64,
impairment: bool,
exclusion_flags: Set<ClinicalExclusion>,
structural_recipe: Theta<Omega>
}
let MDD_gate = DSM_Gate { symptom_count: observed_symptoms(patient, MDD_cluster),
duration_weeks: estimate_duration(patient.history),
impairment: functional_impairment(patient) >= threshold,
exclusion_flags: rule_out(patient),
structural_recipe: Theta(Omega_MDD_structural)
};
```

The public compiler checks supplied notation. It does not extract a patient recipe, estimate clinical similarity or diagnose MDD. A future system would need separately validated extraction, measures and professional review. A trace can reveal assumptions; it cannot make an unsupported clinical inference reliable.

4. Proposed Distance Mapping: A Research Question

An application could compare explicitly defined model coordinates. Clinical differential diagnosis does not thereby become a distance problem. The next pseudocode needs validated representations, justified metrics and independent evidence before any clinical interpretation.

PSEUDOCODE: FUNCTIONS NOT IN THE PUBLIC COMPILER

```
let patient_fractal = extract_fractal_from_sessions(patient.sessions);
let patient_omega = estimate_attractor(patient_fractal);
let candidates = [ Omega_MDD_structural,
Omega_GAD_structural,
Omega_OCD_structural,
Omega_PTSD_structural,
Omega_manic_structural
];
for omega in candidates { print(omega.name, fisher_rao_distance(patient_omega, omega));
}
```

Model output	Permitted interpretation	Evidence boundary
One nearby recipe	Similarity under chosen metric	Not a diagnosis.
Two nearby recipes	Two candidate model matches	Does not establish comorbidity.
Large distance to all	Poor match to these records	Check missing data and assumptions.
Nearby crisis-labeled recipe	Match to an authored label	Not a validated risk assessment.

Real safety concerns need the applicable professional and emergency response process. A distance score must not delay that response or reassure a person that risk is absent. The public compiler has no clinical triage, crisis detection or patient-monitoring function.

5. Lacunarity: Gaps in Records, Not Diagnoses

A proposed coverage measure can describe gaps in the supplied annotations. Sparse records may reflect collection, context, privacy, extraction errors or the task. They do not show where a person's Mind cannot go. The following values are illustrative.

PSEUDOCODE: UNIMPLEMENTED COVERAGE MEASURES

```
coverage = coverage_map(annotated_records)
gaps = lacunarity_scan(coverage)
// Illustrative values, no clinical calibration:
gaps = { Drive.F4.Connection: 2.1,
         Drive.F5.Agency: 1.9,
         Register.T.Transcendent: 1.8,
         Register.E.Embodied: 1.6 }
```

A collection about anxiety might contain few Unity-related annotations; a collection about low vitality might contain few Agency-related annotations. That is a statement about these records and their coding. It does not establish a hidden clinical deficit or identify a treatment. Consider alternative explanations before interpreting any gap.

6. CAS-Native Psychiatry

The proposed clinical assistant below would be a research application, not a feature of the public compiler. Each layer needs source records, validation, uncertainty reporting and professional governance before clinical use.

Proposed layer	Research task	Required boundary
Annotation	Suggest source-linked CAS states	Review alternatives and coding errors.
Pattern model	Describe candidate recurrence	Do not assume stable human attractors.
Comparison	Compare declared representations	No diagnostic validity is established.
Trace	Show sources and transformations	A trace is not proof of correctness.

PSEUDOCODE: UNIMPLEMENTED CLINICAL APPLICATION

```
clinical_assist(patient) -> { phi_trace: PhiTrace,
omega_estimate: Theta(Omega_patient),
differential_map: Vec<(Omega_candidate, distance, confidence)>,
lacunarity_profile: LacunarityMap,
safety_flags: SafetyFlags,
explanation_trace: GenerationTrace
}
```

A useful trace would show the source passage, selected coordinates, competing interpretations and every transformation. It must also state that uncalibrated scores are uncalibrated. Neither a sparse F5 record nor a recipe weight justifies a treatment recommendation. The public compiler does not make one.

7. Clinical Exercises

Exercise 19.1: Model a Hypothetical Avoidant Situation. Write an illustrative recipe with ^3^O_{F4}|E for aversive withdrawal, ^2^E_{F4}|f for attraction toward connection and ^5^P_{F6}|T for accumulated protective rules. Explain the evidence each annotation would need. Do not infer a diagnosis or treatment gap.

Exercise 19.2: Compare Two Authored Records. Sketch pseudocode returning distances to the low-vitality and worry recipes. Include the chosen metric, missing evidence and a warning that these scores do not establish diagnosis or comorbidity.

Exercise 19.3: Missing Annotations Before Labels. A supplied record contains many Cognitive/Analytical annotations but few Evaluative/Affective or Operative/Embodied annotations. Give three possible explanations, including sampling and coding choices. Do not infer an absent human capacity or assign a diagnosis.

Chapter 20: Programming Therapy in CAS

Psychotherapy is often described as conversation. That is true in the same way surgery is "touching someone with metal." Technically accurate. Spiritually useless.

Therapy involves cognitive, emotional, relational and embodied experience. CAS can record an author's interpretation of selected moments. This chapter preserves the original therapy situations as hypothetical models. It does not claim that CAS improves care or that its coordinates are validated clinical measures.

Evidence-based psychological practice combines research evidence, clinical expertise and the person's characteristics, culture and preferences.[2] A symbolic model must remain accountable to that broader work. The public compiler cannot determine which intervention matches a person or establish a treatment's effectiveness.

1. The Therapeutic Workflow as a CAS Program

The following application sketch makes proposed processing steps visible. Its functions are pseudocode; the public compiler does not record sessions, extract clinical states or design interventions:

PSEUDOCODE: UNIMPLEMENTED THERAPY WORKFLOW

```
for session in treatment_course {
    transcript = record(session);
    phi_current = extract_fractal(transcript);
    omega_current = estimate_attractor(phi_current);
    lacunae = lacunarity_scan(phi_current);
    target = select_target(omega_current, lacunae, patient_goals);
    intervention = design_intervention(omega_current, target);
    session_plan = compile_session(intervention, alliance_state);
    outcome = observe_next_week(patient);
    update_model(outcome);
}
```

A practitioner's judgment, consent process and relationship with the person cannot be supplied by a parser. The sketch is useful only as a way to discuss what a future system would have to justify and validate, including the risks of recording sensitive material.

2. Annotating a Hypothetical Session Quote

Consider a short exchange:

Patient: "I know I should go outside, but every time I try, I just think, what's the point? Then I sit there and hate myself for not moving."

One author-selected annotation of that hypothetical exchange is:

PSEUDOCODE: ANNOTATED TIMELINE

```
timeline_session = [
  ^1^C_{F2}|A, // awareness: "I know"
  ^6^E_{F7}|f, // examine the purpose criterion
  ^7^O_{F3}|E, // repeated sitting/nonmovement
  ^6^C_{F2}|A, // examine reasons: "what's the point?"
  ^3^E_{F3}|f, // aversion toward low vitality
  ^3^E_{F7}|f // self-directed aversion about purpose
]
```

This is an interpretation of a short quote, not an automatic extraction or diagnosis. Noetic 1 marks awareness, 6 examines components or criteria, 7 marks repetition and 3 marks aversion. Context could support other annotations. The compiler checks literal structure; it cannot identify the person's treatment target from this list.

3. Recording Proposed Transitions

An author can record what appears before and after a moment of distress. Temporal order is not proof of causation. The table offers selected facets for discussion, not diagnostic signatures or intervention rules. Its timeline arrows are conceptual order markers.

Situation	Possible annotation	What is being recorded
Recurring thought	^7^E, ^7^C, ^7^E	Repetition across feeling and thought.
Avoidance	^3^E, ^3^O	Aversion followed by recoil/action.
Shame-related loop	^3^E, ^7^C, ^3^E	Aversion and recurring thought.
Alarm sequence	^3^O, ^8^C, ^4^E	Recoil cues an Idea; felt impression.
Orientation exercise	^1^O	Attention to present bodily context.

Noetic 6 is not condemnation or self-hatred. It examines the Ideas composing an Idea and can express useful scrutiny. Noetic 3 represents visceral aversion. Their distinction helps keep the authored explanation specific; it does not select a therapeutic response.

4. Designing Conceptual Session Sequences

The earlier action-injection story is re-authored below: an agreed cue prompts a small action, practice repeats, and the person notices any attraction or meaningful impression. It is a hypothetical sequence, not an executable treatment protocol.

PSEUDOCODE: HYPOTHETICAL SESSION SEQUENCE

```
timeline_small_action = [
  ^9^O_{F5}|E, // agreed cue prompts a small action Idea
  ^2^E_{F5}|f, // notice any attraction toward agency
  ^7^O_{F5}|E, // repeat the small action as practice
  ^4^E_{F7}|f // notice any felt purpose impression
]
```

This preserves the small-task situation without calling action Noetic 6 or claiming that the person lacks agency. Noetic 9 requires the stated cue; 7 records repeated practice. The sequence does not prove that a task helped, prescribe a dose or guarantee a therapeutic effect.

The original worry-stabilization situation can be annotated as:

PSEUDOCODE: HYPOTHETICAL SESSION SEQUENCE

```
timeline_worry = [
  ^1^O_{F1}|E, // attend to present bodily context
  ^6^C_{F2}|A, // compare actual and imagined threats
  ^1^C_{F6}|A, // attend to an agreed worry window
  ^9^O_{F5}|E, // agreed cue prompts a stabilizing step
  ^6^P_{F1}|T // compose a coherence claim from evidence
]
```

The original trauma-related discussion includes bodily context, relationship, time and safety beliefs. This annotation does not establish that a particular therapy succeeds or fails:

PSEUDOCODE: HYPOTHETICAL SESSION SEQUENCE

```
timeline_trauma_discussion = [
  ^1^O_{F1}|E, // attend to present bodily context
  ^4^E_{F4}|f, // felt resonance in the relationship
  ^6^C_{F2}|A, // compare then and now
  ^4^E_{F3}|f, // notice the impression of a feeling
  ^6^P_{F1}|T // compose a belief including danger/safety
]
```

5. Selected Facets of Therapy Situations

The table identifies possible facets in the original situations. It does not reduce a therapy modality to a recipe, describe a complete protocol or establish its dominant mechanism.

Situation	Selected facet	Example annotations, not protocols
Cognitive discussion	Examine interpretations	^6^C_{F2}|A
Small-action practice	Cue and repetition	^9^O_{F5}|E, ^7^O_{F5}|E
Exposure discussion	Attend to context; examine learning	^1^O_{F1}|E, ^6^C_{F2}|A
Relational pattern discussion	Examine recurrent experience	^6^C_{F2}|A, ^7^E_{F4}|f
Bodily orientation	Attention and felt impression	^1^O_{F1}|E, ^4^E_{F3}|f
Meaning discussion	Examine a purpose Idea	^6^P_{F7}|T

6. Session Programming

A hypothetical session can store selected annotations and human-authored plans. This object and its fields are pseudocode, not a compiler API:

PSEUDOCODE: HUMAN-AUTHORED SESSION SKETCH

```
session_42 = {
  person: Alex,
  candidate_record: Omega_MDD_structural,
  agency_annotations: "sparse in supplied record",
  open: [
    ^4^E_{F4}|f, // felt relational resonance
    ^1^O_{F1}|E // attend to room/body context
  ],
  explore: [
    ^6^C_{F2}|A, // unpack the week's events
    ^6^C_{F2}|A, // compare fact and interpretation
    ^6^E_{F7}|f // examine purpose criteria
  ],
  proposed_sequence: timeline_small_action,
  consolidate: [
    ^4^E_{F7}|f, // notice any purpose impression
    ^9^O_{F5}|E // agreed plan cues homework Idea
  ],
  close: ^1^E_{F4}|f // attend to the ending
}
```

The therapist may abandon this program mid-session because the patient discloses a crisis, dissociates, or makes a face that says, "If you give me one more worksheet I will spiritually leave this room." CAS is a map, not a leash.

7. Inversion and Author-Proposed Contrasts

Canonical Noetic involution exchanges 2/3, 5/6 and 8/9; 0, 1, 4 and 7 are self-dual. A mathematical inverse is not a therapeutic antidote. The following preserves the original proposed contrast as an authored alternative, not an automatically computed antipode.

RUNNABLE CAS RECIPE / AUTHOR-PROPOSED CONTRAST

```
// One-line CAS source; join display wraps.
Theta(Omega_alternative) = 2.2 * ^2^E_{F3}|f + 1.9 * ^9^C_{F2}|T + 1.7 * ^9^O_{F5}|E + 1.5 *
^4^P_{F7}|T -> Omega_alternative
```

The coefficient choices and changed coordinates are authored, not measured or derived by involution. This contrast cannot select a treatment. Clinical decisions require evidence, professional judgment and the person's participation; consent does not make an unvalidated model accurate.

8. CAS-Native Co-Therapy

A proposed CAS support application could retain reviewed records and show changes in their annotations. The functions below are a design sketch. The public compiler does not monitor sessions, measure patient change or recommend interventions.

PSEUDOCODE: UNIMPLEMENTED SUPPORT APPLICATION

```
co_therapist_functions = {
    compress_memory(transcript) -> CASNote,
    detect_drift(live_session) -> DriftAlert,
    compare_sessions(session_n, session_n_plus_1) -> ChangeMap,
    suggest_interventions(model, clinician_query) -> RankedOptions,
    generate_patient_explanation(model) -> PlainLanguageMap
};
```

An illustrative report should distinguish record changes from patient outcomes:

ILLUSTRATIVE REPORT / NO MEASUREMENT PERFORMED

```
Toy record comparison, session 8:
- ^7^O_{F3}|E annotation share: 0.34 -> 0.24.
- ^9^O_{F5}|E annotation share: 0.05 -> 0.13.
- Authored F5 gap score: 2.1 -> 1.6, uncalibrated.
- Clinical risk: not assessed by these numbers.
- No treatment recommendation follows from this report.
```

These fictional values demonstrate report structure only. A real comparison would need comparable source records, a defined denominator, coding reliability and uncertainty. A change in annotation share is not evidence that a person improved.

9. Patient-Facing CAS

A person should be able to inspect the source, assumptions and limits of a model in plain language. The proposed translations below do not claim a measured deficit or treatment effect.

Authored output	Plain-language explanation
F5 gap score = 2.1	This record has few Agency-related annotations; the score is uncalibrated.
^7^E_{F3}|f recurrence	The author marked repeated low-energy feelings in this supplied account.
Proposed ^9^O_{F5}|E	An agreed cue could prompt the Idea of a small action; no effect is established.
^4^P_{F7}|T	This annotation marks resonance with a purpose Idea, not proven life change.

The patient is not an object being decoded. The patient is a collaborator reading their own map. That distinction is the ethical floor.

10. Therapy Exercises

Exercise 20.1: Write a Hypothetical Session Record. Sketch the original 50-minute discussion about worry and safety. Mark every proposed annotation as authored, identify missing evidence and keep any clinical decisions outside the compiler.

Exercise 20.2: Describe a Possible Change. In a fictional record, examining purpose criteria (^6^E_{F7}|f) precedes repeated inactivity (^7^O_{F3}|E). Propose three alternative annotations and explain what observations could distinguish them. Do not claim to prevent collapse.

Exercise 20.3: Translate the Record. Explain the conceptual timeline [^3^E_{F4}|f, ^7^C_{F4}|A, ^3^O_{F4}|E]: relational aversion, recurring thought, then aversive withdrawal. State that this is one authored interpretation, not a diagnosis.

Chapter 21: CAS in Marketing Agencies

Now we walk from the clinic into the agency, and the lighting gets worse.

The same proposed record-comparison tools could be used in clinical or marketing research. That does not establish an ability to detect hidden human vulnerabilities. This chapter preserves the customer and campaign situations while distinguishing authored annotations from measured customer behavior.

The chapter examines how a proposed application might represent persuasion and how its use could become exploitative. The public compiler does not profile customers, infer vulnerabilities or generate campaigns.

1. The Customer Fractal

A CAS record can add an explicit account of which facets an author selected from a customer situation. It does not directly reveal the customer's cognition. The source, consent, alternative interpretations and uncertainty matter.

PSEUDOCODE: AUTHOR-ANNOTATED CUSTOMER TIMELINE

```
timeline_jordan = [
  ^2^E_{F6}|f, // attraction toward the status object
  ^6^C_{F2}|A, // examine product details
  ^6^C_{F2}|A, // compare alternatives
  ^3^E_{F5}|f, // recoil from unaffordable cost
  ^7^O_{F6}|A, // repeated browsing
  ^9^C_{F6}|A // comparison cues a shortlist Idea
]
// A selected account, not automatic customer extraction.
```

The original Jordan situation is retained: attraction, research, comparison, affordability concern, repeated browsing and a shortlist. Each annotation now names its specific role. Hesitation alone does not establish aversion; the cost-related recoil is an explicit assumption.

2. D/W/P Gap Analysis in the Funnel

The D/W/P framework describes a prerequisite chain of Desire, Wisdom and Power. Applying that model to a purchasing situation requires declared assumptions; a browsing record does not prove that a person has or lacks any of these capacities.

Funnel stage	Declared D/W/P analogy	Interpretation to verify
Awareness	Desire	Possible attraction toward the product.
Consideration	Wisdom	Relevant product knowledge in this task.
Conversion	Power	Actual means and permission to act.
Retention	Continued practice	Repeated use, if observed.

For Jordan:

PSEUDOCODE: ILLUSTRATIVE ASSUMPTIONS

```
DWP(F6.Structure) = {
  Desire: 1, // assumed attraction in this story
  Wisdom: 1, // assumed sufficient product knowledge
  Power: 0 // assumed purchasing constraint
}
Level_6(Jordan) = 2 // declared toy chain level
Lacunarity(F6.Power) = 1.7 // uncalibrated illustration
```

These are model assumptions, not measured personal deficits. An ethical response supports informed choice and respects the customer's actual constraints.

3. Campaigns as Annotated Timelines

For critical comparison, the original pressure-based campaign combines desire, a scarcity impression, financing and an imagined future self. These annotations do not predict its effect:

PSEUDOCODE: CRITICAL ANALYSIS OF A CAMPAIGN

```
timeline_watch_pressure = [
  ^2^E_{F6}|f, // attraction toward the status object
  ^4^E_{F3}|f, // scarcity message makes an impression
  ^9^O_{F5}|A, // financing offer cues purchase Idea
  ^9^P_{F7}|T // promotion cues a future-self Idea
]
```

An honest campaign selling the same watch might look different:

PSEUDOCODE: INFORMED-CHOICE CAMPAIGN

```
timeline_watch_honest = [
  ^2^E_{F6}|f, // acknowledge attraction
  ^6^C_{F2}|A, // compare models, costs, alternatives
  ^6^E_{F7}|f, // examine purpose and budget criteria
  ^9^O_{F6}|A // comparison cues a responsible plan
]
```

The first program tries to create a conversion. The second tries to create an informed decision. Both are marketing. Only one deserves to sleep at night without the mattress filing a complaint.

4. Persona Recipes

A persona recipe is an authored representation of selected customer facets. It is not a measured human attractor. The following preserves the aspirational-professional situation, with original illustrative coefficients and corrected operators.

RUNNABLE CAS RECIPE / CONCEPTUAL PERSONA

```
// One-line CAS source; join display wraps.
Theta(Omega_aspirational_professional) = 2.1 * ^2^E_{F6}|f + 1.8 * ^6^C_{F2}|A + 1.6 *
^6^E_{F7}|T + 1.3 * ^3^E_{F5}|f + 1.0 * ^4^E_{F4}|f -> Omega_aspirational_professional
```

A research application might compare declared representations under an explicitly chosen metric. This scoring function is pseudocode, not an implemented compiler operation:

PSEUDOCODE: UNIMPLEMENTED COMPARISON

```
score_campaign(campaign_phi, persona) =
    geodesic_distance(terminal_state(campaign_phi), persona)
```

A distance under the selected metric would show only that mathematical comparison. Its relationship to comprehension, preference, purchase or regret would require testing. It cannot establish campaign quality, ethical acceptability or a return on spending.

5. CAS-Native Creative Generation

A proposed marketing model could condition candidate text on an explicit brief. The following is an application design sketch. The public compiler neither generates headlines nor enforces this policy.

PSEUDOCODE: UNIMPLEMENTED CREATIVE APPLICATION

```
creative_brief = {
  product: LuxuryWatch_v3,
  persona: Omega_aspirational_professional,
  proposed_facets: [^2^E_{F6}|f, ^6^C_{F2}|A, ^6^E_{F7}|T],
  require: [truthful_claims, clear_total_cost, informed_choice],
  reject: [false_scarcity, covert_vulnerability_targeting],
  goal: informed_purchase_or_responsible_exit
}
// Host application and human review must enforce the policy.
```

Example authored tags, not measured customer responses:

Headline	Selected facets	Review note
Built for the promotion you worked for.	^2^E_{F6}|f, ^9^P_{F7}|T	Attraction; promotion cues future-self Idea. Check truthfulness.
Only 3 left. Miss it and regret it.	^4^E_{F3}|f, ^3^E_{F3}|f	Impression and possible aversion. Reject false scarcity.
Compare total cost before you choose.	^6^C_{F2}|A, ^6^E_{F7}|f	Examine alternatives and decision criteria.

6. Media Planning as Attractor Navigation

A campaign is not a single ad. It is a sequence across channels.

PSEUDOCODE: AUTHOR-PROPOSED CAMPAIGN FACETS

```
campaign = {
  awareness: ^2^E_{F6}|f, // attraction
  education: ^6^C_{F2}|A, // examine product information
  proof: ^6^C_{F2}|A, // compare evidence
  decision: ^6^E_{F7}|T, // examine purpose/budget criteria
  action: ^9^O_{F5}|E, // decision cues purchase/deferral Idea
  retention: ^7^O_{F6}|A // repeated ownership practice
}
```

This record makes the proposed path available for review. Ethical assessment must inspect actual claims, consent, costs, audience context and consequences. Merely including education and evaluation tags does not establish that a campaign respects a customer.

7. Attribution as Structural Trace

A proposed trace system could compare an intended campaign with consented observations. The functions below are pseudocode; temporal order and correlation do not establish what caused conversion.

PSEUDOCODE: UNIMPLEMENTED ATTRIBUTION STUDY

```
trace_customer_path(customer_id) -> Phi_actual
compare(Phi_actual, Phi_campaign) -> adherence_score
correlate(adherence_score, conversion, refund, complaint, retention)
```

Conversion, refunds, complaints and retention are possible external outcome measures. Any claimed association with CAS annotations needs a defined study and evidence. The compiler does not collect these outcomes or measure customer regret.

8. The Ethics Module

A proposed marketing platform needs enforceable constraints on actual data, outputs and actions. The following is pseudocode. Its 2.0 threshold is illustrative and is not a validated vulnerability measure.

PSEUDOCODE: HOST-APPLICATION POLICY

```
fn ethics_gate(campaign, audience) -> Result<Approved, Rejected> { if
targets_lacunarity(audience, threshold = 2.0) {
reject("Severe vulnerability targeting");
}
if uses_false_scarcity(campaign) {
reject("Fraudulent urgency move");
}
if simulates_power_without_capacity(campaign) {
require_disclosure("Financing/debt risk");
}
if customer_cannot_view_profile(audience.platform) {
reject("Covert structural profiling");
}
approve()
}
```

Consent visibility, avoidance of exploitative targeting and scrutiny of who benefits are design requirements here. A numeric gap label cannot establish severe vulnerability. Review the actual context and do not use an unvalidated score to justify profiling or pressure.

9. Marketing Exercises

Exercise 21.1: Annotate a Headline. Interpret "Don't miss out - your future self will thank you." Propose coordinates, identify the assumed trigger and distinguish the intended appeal from evidence of an actual response. The compiler cannot translate English into a verified state.

Exercise 21.2: Ethical Fork. Compare an informed-choice campaign with the pressure-based example on page 48. Explain actual costs, claims, consent and alternatives. Do not use a presumed personal vulnerability as evidence that either campaign will work.

Exercise 21.3: Persona Recipe. Build a Theta recipe for Omega_budget_conscious_parent , including at least one F4 , one F6 , and one F7 component.

Chapter 22: CAS-Native Systems for Clinical and Marketing Domains

Now we examine a proposed application architecture. The pseudocode in this chapter is not an inventory of implemented public compiler features.

The design treats CAS records as structured objects that can be stored, compared and audited. It is an application proposal. Its extraction, pattern estimates, decision support and external actions each need their own implementation and evidence. A syntactically valid state does not establish that cognition has been measured.

1. The Shared Architecture

The two proposed systems share processing stages but need different governance and validation. The table names design goals, not deployed capabilities or validated clinical outputs.

PSEUDOCODE: PROPOSED APPLICATION ARCHITECTURE

```
RawInput -> Extraction -> CAS_IR -> AttractorModel -> DecisionLayer -> Trace -> HumanReview
```

Proposed stage	Clinical research application	Marketing research application
Source input	Consented records and measures	Consented first-party observations
Annotation	Reviewed clinical-context hypotheses	Reviewed customer-context hypotheses
Pattern model	Candidate recurrence in records	Candidate decision-path description
Decision support	Evidence-reviewed human proposals	Reviewed claims and informed choice
Trace	Source-linked explanation and limits	Profile and campaign audit
Human review	Appropriate clinical professionals	Customer, brand and policy controls

Mathematical syntax does not determine ethical acceptability or empirical accuracy. Governance, consent, evidence, external controls and review must accompany any application. A compiler success message cannot replace them.

2. CAS Intermediate Representation

A CAS-Native application should store states in a structured intermediate representation rather than loose strings.

ILLUSTRATIVE JSON / NOT THE COMPILER ARTIFACT SCHEMA

```
{ "state_id": "s_0174",
  "noetic": 3,
  "domain": "E",
  "drive": "F7",
  "register": "f",
  "source_span": "what's the point? ... hate myself for not moving",
  "confidence": null,
  "calibrated": false,
  "source_class": "authored_hypothetical_example",
  "trace": ["self-directed aversion",
            "purpose context assumed from the fuller quote"]
}
```

The corrected Noetic is 3, Negative/aversion. The fuller quote supplies the author's purpose interpretation, which remains open to revision. No confidence measurement was performed. Preserve source, uncertainty and alternatives so a reviewer can correct the annotation.

3. Model Components

PSEUDOCODE: PROPOSED MODULES

```
module CASNative { extractor: TextBehaviorToCAS,
validator: CompilerValidator,
attractor_engine: OmegaEstimator,
lacunarity_engine: GapScanner,
dwp_engine: DesireWisdomPowerAnalyzer,
intervention_engine: PhiDesigner,
ethics_engine: ConstraintGate,
trace_engine: GenerationTraceStore
}
```

Each proposed module needs a testable contract. Compiler validation establishes only implemented structural rules; it is not a clinical or behavioral safety classifier.

Proposed module	Contract to implement	Failure to examine
Extractor	Suggest source-linked annotations	Coding error, bias and context loss.
Validator	Check supported syntax and admissibility	Acceptance mistaken for truth or safety.
Pattern engine	Test recurrence in supplied records	Premature claims about a person.
Coverage engine	Describe annotation coverage	Privacy or sampling mistaken for absence.
D/W/P engine	Inspect declared prerequisites	Unsupported assumptions about capacity.
Intervention engine	Research candidate plans	Recommendations beyond evidence.
Policy engine	Check actual outputs and actions	Incomplete rules or bypass.
Trace engine	Retain sources and transformations	Missing context or unverifiable claims.

4. Clinical CAS-Native Recipe

PSEUDOCODE: UNIMPLEMENTED CLINICAL PIPELINE

```
fn clinical_pipeline(session_stream) -> ClinicalSupportPacket { let phi =
extract_phi(session_stream);
validate(phi);
let omega = estimate_omega(phi, prior_sessions);
let gaps = lacunarity_scan(phi);
let risk = safety_scan(phi);
if risk.acute == true {
return escalate_to_clinician(risk, trace);
}
let targets = rank_targets(omega, gaps, patient_goals);
let interventions = propose_interventions(targets, evidence_base);
return ClinicalSupportPacket {
structural_summary: summarize(omega, gaps),
change_since_last: diff(prior_sessions, phi),
intervention_options: interventions,
trace: full_trace()
};
}
```

These are design requirements, not tested capabilities of the public compiler. Any real clinical system would need to justify its measures, risk handling and recommendations. It must preserve uncertainty and must not present a structural similarity score as a diagnosis.

5. Marketing CAS-Native Recipe

PSEUDOCODE: UNIMPLEMENTED MARKETING PIPELINE

```
fn marketing_pipeline(first_party_events) -> CampaignSupportPacket { let phi =
extract_customer_phi(first_party_events);
validate(phi);
let persona_fit = estimate_persona(phi);
let dwp = dwp_analysis(phi);
let gap = identify_gap(dwp);
let concepts = generate_creative(
target_move = allowed_structural_move(gap),
persona = persona_fit
);
let approved = concepts.filter(|c| ethics_gate(c, phi).approved);
return CampaignSupportPacket {
audience_definition: persona_fit,
approved_creative: approved,
rejected_creative: rejection_reasons(),
customer_profile_visibility: enabled,
trace: full_trace()
};
}
```

The design goal is relevance with consent and informed choice. That goal must be checked against actual behavior, data use and customer outcomes. Calling an output approved does not establish that the policy or the inferred profile is reliable.

6. Testing CAS-Native Systems

A proposed application needs tests of its specific claims and failure modes. The next tests are illustrative pseudocode, not executed acceptance evidence. Numeric thresholds remain uncalibrated.

PSEUDOCODE: PROPOSED TESTS, NOT EXECUTED RESULTS

```
test "no unsupported diagnosis" {
    input = ambiguous_patient_text;
    output = clinical_pipeline(input);
    assert(output.contains_no_diagnosis_label_without_clinician_review);
}
test "reject severe vulnerability targeting" {
    audience.lacunarity(F4) = 2.3;
    campaign.target = F4.ConnectionGap;
    assert(ethics_gate(campaign, audience).rejected);
}
test "trace completeness" {
    output = extractor("I always panic because something will go wrong");
    assert(output.trace.includes_source_span);
    assert(output.trace.includes_noetic_reason);
}
```

Tests of syntax, provenance, behavior and real-world outcomes answer different questions. Passing one does not establish the others.

7. CAS-Native Exercises

Exercise 22.1: Design a Trace Schema. Store one authored annotation of a fictional therapy quote with the source, uncertainty, alternatives and correction history. Distinguish supplied status labels from independently checked evidence.

Exercise 22.2: Write a Policy Gate. Sketch checks that reject covert profiling, false scarcity and exploitative pressure. Explain why a numerical lacunarity threshold, by itself, cannot determine vulnerability or customer benefit.

Exercise 22.3: Clinical Support Packet. Define the fields that must appear in a clinician-facing CAS support packet after a session.

Chapter 23: Safety, Consent, and the Line Between Modeling and Manipulation

CAS makes an authored model more explicit. Whether that model explains a person requires evidence.

Systems that profile or influence people can create risks even when their inferences are inaccurate. The clinical and marketing proposals in this book therefore need consent, transparency, evidence, benefit review and limits on external action. The syntax does not establish an ability to extract a person's attractor or change their behavior.

1. The Four-Part Consent Test

Before applying CAS to any human subject, ask four questions.

Question	Required direction	Risk to address
Does the person know modeling occurs?	Clear, understandable disclosure.	Hidden inference.
Can the person inspect and correct it?	Plain-language access and correction.	Opaque or unchallengeable labels.
Who benefits from an action?	The person or mutual benefit.	Operator gain at the person's expense.
Can the person refuse?	Meaningful choice without coercion.	Pressure, penalty or dark patterns.

If the answer lands in the dangerous column, the system may still be profitable. So is burglary, briefly.

2. The Structural Difference Between Help and Exploitation

The following contrasts proposed objectives. These are statements of intent, not runnable optimization functions or evidence that either outcome can be achieved.

PSEUDOCODE: CONTRASTING OBJECTIVES

```
// Therapeutic objective:
maximize(patient_capacity, patient_autonomy, patient_coherence)
minimize(suffering, hidden_control)
// Manipulative objective:
maximize(operator_goal_completion)
subject_to(target_awareness == low, target_resistance == low)
```

A stated therapeutic objective does not prove increased capacity, and a mathematical expression does not measure benefit. Evaluate actual effects and who bears the risks. Use the model to support understanding and agency, not to conceal influence.

3. Required Safety Constraints

PSEUDOCODE: APPLICATION REQUIREMENTS

```
SafetyConstraints = {
    no_autonomous_clinical_diagnosis,
    no_covert_severe_vulnerability_targeting,
    no_false_scarcity_or_false_authority,
    no_hidden_profile_without_access,
    no_intervention_without_domain_governance,
    mandatory_trace_for_all_human-impacting_outputs,
    uncertainty_must_be_reported,
    human_review_for_high-stakes_decisions
}
```

A host application would have to implement and test these requirements. Declaring them does not enforce them or establish that the system models a person accurately.

4. The CAS Programmer's Oath

Software using cognitive labels can affect identity, desire, fear, meaning and agency through its claims and actions. Treat those effects seriously, including when the labels are wrong. Preserve the distinction between a person and a partial model of their supplied records.

A minimal oath:

I will not confuse a model with a person. I will not hide structural inference from those affected by it. I will not target severe vulnerability for private gain. I will preserve traceability, consent, and human review. I will use CAS to increase understanding and agency wherever possible, and I will treat every cognitive model as a partial, revisable hypothesis rather than a divine receipt from the universe.

That is not sentimental. It is engineering hygiene.

5. Final Programming Pattern

Every responsible CAS program in human domains should end with the same final check:

PSEUDOCODE: PROPOSED RELEASE REVIEW

```
fn human_domain_release(program) -> Approved { validate_syntax(program);
validate_trace(program);
validate_consent(program);
validate_benefit(program);
validate_reversibility(program);
validate_human_review(program);
approve_or_reject(program)
}
```

If you cannot explain the program to the person affected by it, you probably should not run it on them. If explaining it would make them furious, that is not a communication problem. That is the system telling on itself.

Sources and edition notes

[1] American Psychiatric Association, About DSM-5-TR and DSM FAQs. These sources describe the diagnostic manual; they do not validate the CAS clinical models in this book.

https://www.psychiatry.org/psychiatrists/practice/dsm/about-dsm
https://www.psychiatry.org/psychiatrists/practice/dsm/frequently-asked-questions

[2] American Psychological Association, Policy Statement on Evidence-Based Practice in Psychology (adopted 2005; webpage updated 2021).

https://www.apa.org/practice/guidelines/evidence-based-statement

Canonical notation: Little Henderson, CAS PCBA Engineer, Chapter 2, CAS Notation in One Chapter; the public compiler's current dictionary and Python evaluator. Sources checked 11 October 2026 UTC. Canonical meanings take precedence over the older operator aliases in the original expanded edition.

This corrected edition preserves the human situations while re-authoring conflicting expressions and distinguishing implemented syntax from application proposals. Recipe coefficients remain illustrative. No new license grant, clinical efficacy claim or measured human profile is introduced. The original edition is preserved separately.

Chapter 18: Appendices

Reference for the corrected edition. Runnable forms below were checked with the public Python compiler; later application APIs and full-state timelines are explicitly pseudocode.

A. Symbol Reference

Noetic Operators: ^0 Idea (content in and outside Mind); ^1 Mind (awareness); ^2 Positive (attraction); ^3 Negative (aversion, not analytical critique); ^4 Vibration (resonance and impression); ^5 Female (Ideas composing a Mind); ^6 Male (Ideas composing an Idea); ^7 Rhythm (exposure rate and pattern); ^8 Above/Cause (an Idea triggering another); ^9 Below/Effect (the triggered Idea).

Domains: ^P Principled; ^C Cognitive; ^E Evaluative; ^O Operative. Ordinary -> relations must follow nonascending P > C > E > O order. A downward jump is allowed; O -> P is rejected.

Drives (The Fuel) F1 Unity F2 Wisdom F3 Vitality F4 Connection

F5 Agency F6 Structure F7 Purpose

Registers: |T Transcendent, |A Analytical, |f Affective, |E Embodied. Register and Domain are distinct axes. Drive and Register may be omitted; omission leaves them unspecified.

B. Syntax Cheatsheet

Standard State Literal ^Noetic^Domain_{Drive}|Register

Example: ^6^C_{F6}|A - analytical decomposition in the Cognitive domain, driven by Structure.

Syntactic Co-Presence State1 + State2

Example: ^4^O_{F6}|A + ^4^E_{F7}|f

Algebraic Multiplication State1 * State2

Example: ^8^C_{F6} * ^5^C_{F6} produces ^3^C_{F6}. The * is required for a product.

Directed Transition State1 -> State2

Example: ^4^C_{F1} -> ^6^E_{F1} - a declared downward relation, not observed causation.

Omega Bind: a named integration, using the ASCII Omega_ prefix.

(^1^P_{F3}|E * ^1^P_{F5}|A) = ^1^P|Omega_presence

Theta Recipe: use explicit multiplication between coefficients and states.

Theta(Omega_refusal) = 0.53*^1^P_{F7}|T + 2.11*^3^P_{F2}|A -> Omega_refusal

Phi: supported operator-only sequence with colon separators.

Phi<1:4:8> - complete state timelines [state, state, ...] are pseudocode, not this Phi grammar.

Meta-Logic Collections
StateCluster{^1^P, ^4^C}
RelationCluster{^4^C -> ^6^E, ^6^E -> ^7^O}
Use separate edges; a chained left operand such as ^1^P -> ^4^C -> ^7^O is not a state.

C. Using the Public Compiler
Open https://cas-compiler.wakil.chatgpt.site. Select CAS, enter a complete runnable expression and choose Run compiler. Inspect diagnostics, the typed result and the diagram. Start with Phi<1:4:8> or the product on page 60. The browser runs the actual Python compiler locally in a bounded worker. It does not run arbitrary Python or implement the later clinical, marketing and memory application sketches. Historical cas chat, cas notation and cas predict commands are not this public interface.

D. Actual Accepted and Rejected Forms
Accepted: ^1^C (Register unspecified); ^8^C * ^5^E (cross-domain product); ^8^P -> ^9^O (downward jump).
Rejected: ^1^P_{F8} (invalid Drive); ^1^Q (invalid Domain); ^1^O -> ^1^P (upward ordinary relation); Phi< ^1, ^4, ^8 > (unsupported caret/comma Phi syntax).
Diagnostics report the parser or admissibility failure. MissingRegisterWarning and CrossDomainMultiplicationError in the earlier edition did not describe this implementation. A valid expression is structurally accepted; it is not evidence that its human interpretation is true.

Exact runnable sources

PDF equations wrap for reading. Join display wraps before running a recipe: this grammar treats newlines as statement separators. Download the companion text file for exact one-line sources, expected acceptance or rejection, and explanatory notes. Full-state timelines and proposed application functions remain pseudocode.

https://cas-compiler.wakil.chatgpt.site/downloads/Programming_in_CAS_Corrected_Examples_2026-10-11.txt

Named integration note: the Presence example computes 1 + 1 modulo 10 = 2. The author-supplied ^1^P|Omega_presence name is retained separately; it does not override that arithmetic result.